SidewalkPilot Steering Model Report

Dataset-specific current-label offline evaluation

Generated 2026-07-13 15:09:00. Series 1/2 checkpoints are evaluated on their original 2,224-image corrected real field dataset. Series 3 checkpoints are evaluated on the current 81,237-image Series 3 field dataset. Rankings and metric gradients are therefore separated by evaluation dataset; values must not be used for direct cross-dataset ranking. Checkpoint naming uses X.Y for the final epoch and X.Yb for the best epoch.

Series 1/2 class-balanced leader: 2.3b, with Bal9 79.2%, turn exact 78.8%, and turn +/-1 98.4%. Lowest Series 1/2 MAE: 2.4b at 3.265 deg.

Series 3 class-balanced leader: 3.2, with Bal9 34.0%, turn exact 31.1%, and turn +/-1 57.7%. Lowest Series 3 MAE: 3.4b at 13.904 deg.

Series 1 uses raw BGR and output scale 86. Series 2 uses output scale 85; v2.0/v2.0b use legacy HSV/CLAHE preprocessing, while v2.1 and newer use raw BGR. Series 1/2 produce one continuous steering value. Series 3.0 and 3.0b produce two continuous values (steering and throttle). Series 3.1 and newer produce 19 raw values: 9 steering-class logits, 9 within-class offsets, and throttle. Every architecture is decoded to steering degrees before the common steering metrics are calculated.

Series 1/2 Evaluation Sources

Source	Count	Purpose
D0328_17	315	First dataset relabel, March 28
D0329_15	413	First dataset relabel, March 29
D0425_14	65	Street test
D0426_18	53	Curves and shadows
D0427_18	72	Curved curb
D0429_19	53	Driveway and shadow
D0502_12	154	Shadow fix set
D0502_19	156	Hard turns, curb hugging, smoothness
D0503_17	159	Harsh sidewalk
D0506_20	24	8pm sidewalk failure set
D0510_18	167	v2.3 field-failure run 1: turns, road-right driving, driveways
D0510_19	8	v2.3 field-failure run 2 (short)
D0510_20	585	v2.3 field-failure run 3: turns, road-right driving, driveways

Series 3 Evaluation Sources

Source	Count	Purpose
D0702_16	50684	Series 3 July 2 base set: two manual field runs, 50,684 images
D0707_16	6524	Bright-sun HARD/diagonal shadows across the sidewalk (v3.2 shadow-robustness batch)
D0712_15	2102	Series 3 manual field collection, July 12 run 1
D0712_16	21927	Series 3 manual field collection, July 12 run 2 (103-frame bad segment removed)

Series 1/2 Class-Balanced Model Ranking

Ranked by macro-average exact recall across the 9 steering buckets, then turn within-one-bucket recall, then MAE. Each metric column has its own red-yellow-green scale: green means better within this table. Signed error is greenest nearest zero.

Rank	Model	Checkpoint filename	Prep	Bal9	Turn exact	Turn +/-1	ST exact	MAE	Med	Signed
1	2.3b	SidewalkPilot-v2.3b.pth	raw BGR	79.2%	78.8%	98.4%	62.8%	3.41	2.05	0.41
2	2.4b	SidewalkPilot-v2.4b.pth	raw BGR	78.2%	75.9%	99.5%	55.0%	3.27	2.39	-0.04
3	2.3	SidewalkPilot-v2.3.pth	raw BGR	77.8%	76.6%	98.5%	63.0%	3.30	1.93	0.01
4	2.4	SidewalkPilot-v2.4.pth	raw BGR	77.5%	74.9%	99.5%	55.5%	3.28	2.46	-0.23
5	2.2	SidewalkPilot-v2.2.pth	raw BGR	64.6%	62.4%	92.4%	42.3%	6.68	4.26	-0.22
6	2.2b	SidewalkPilot-v2.2b.pth	raw BGR	64.3%	62.1%	92.1%	42.3%	6.68	4.23	-0.46
7	2.1	SidewalkPilot-v2.1.pth	raw BGR	54.2%	51.0%	79.4%	39.4%	10.80	5.83	-1.15
8	2.0b	SidewalkPilot-v2.0b.pth	HSV/CLAHE -> BGR	54.2%	50.0%	78.8%	38.9%	11.13	5.07	-2.62
9	2.1b	SidewalkPilot-v2.1b.pth	raw BGR	54.1%	50.9%	79.6%	38.2%	10.75	5.80	-1.12
10	2.0	SidewalkPilot-v2.0.pth	HSV/CLAHE -> BGR	53.8%	49.5%	78.9%	38.5%	11.12	5.09	-2.63
11	1.9b	SidewalkPilot-v1.9b.pth	raw BGR	43.2%	41.8%	75.3%	37.3%	12.94	6.60	-2.06
12	1.9	SidewalkPilot-v1.9.pth	raw BGR	43.1%	41.3%	75.1%	36.9%	12.98	6.63	-2.06
13	1.8b	SidewalkPilot-v1.8b.pth	raw BGR	38.7%	35.8%	69.6%	36.0%	15.41	9.36	-0.70
14	1.8	SidewalkPilot-v1.8.pth	raw BGR	38.6%	35.8%	69.5%	36.1%	15.47	9.46	-0.65
15	1.7b	SidewalkPilot-v1.7b.pth	raw BGR	35.4%	35.1%	69.9%	37.5%	15.54	9.53	-0.87
16	1.7	SidewalkPilot-v1.7.pth	raw BGR	35.0%	34.9%	69.2%	37.3%	15.82	9.66	-0.48
17	1.5	SidewalkPilot-v1.5.pth	raw BGR	30.8%	30.7%	63.9%	33.3%	16.82	10.59	-0.89
18	1.5b	SidewalkPilot-v1.5b.pth	raw BGR	30.8%	30.7%	63.9%	33.3%	16.82	10.59	-0.89
19	1.4	SidewalkPilot-v1.4.pth	raw BGR	30.5%	29.4%	63.3%	36.6%	16.81	10.19	-3.99

Rank	Model	Checkpoint filename	Prep	Bal9	Turn exact	Turn +/-1	ST exact	MAE	Med	Signed
20	1.6b	SidewalkPilot-v1.6b.pth	raw BGR	30.0%	30.5%	64.9%	30.5%	17.41	11.04	-0.01
21	1.6	SidewalkPilot-v1.6.pth	raw BGR	29.6%	29.6%	64.7%	29.9%	17.53	11.21	0.21
22	1.4b	SidewalkPilot-v1.4b.pth	raw BGR	29.0%	28.5%	62.2%	40.0%	17.04	10.07	-4.15
23	1.2	SidewalkPilot-v1.2.pth	raw BGR	27.8%	27.4%	61.5%	32.9%	18.64	11.79	-3.55
24	1.2b	SidewalkPilot-v1.2b.pth	raw BGR	27.4%	26.6%	62.4%	31.6%	18.63	11.85	-3.73
25	1.3	SidewalkPilot-v1.3.pth	raw BGR	27.0%	26.2%	62.5%	30.4%	18.55	11.81	-3.43
26	1.3b	SidewalkPilot-v1.3b.pth	raw BGR	26.9%	27.0%	62.5%	32.2%	18.48	11.24	-3.65
27	1.1	SidewalkPilot-v1.1.pth	raw BGR	24.7%	24.1%	56.1%	26.1%	20.85	13.42	-5.16
28	1.1b	SidewalkPilot-v1.1b.pth	raw BGR	24.7%	24.1%	56.1%	26.1%	20.85	13.42	-5.16
29	1.0b	SidewalkPilot-v1.0b.pth	raw BGR	17.6%	17.1%	47.1%	20.6%	25.07	17.45	-7.08
30	1.0	SidewalkPilot-v1.0.pth	raw BGR	15.3%	15.5%	45.1%	18.9%	25.55	18.46	-7.45

Series 3 Class-Balanced Model Ranking

Ranked by macro-average exact recall across the 9 steering buckets, then turn within-one-bucket recall, then MAE. Each metric column has its own red-yellow-green scale: green means better within this table. Signed error is greenest nearest zero.

Rank	Model	Checkpoint filename	Prep	Bal9	Turn exact	Turn +/-1	ST exact	MAE	Med	Signed
1	3.2	SidewalkPilot-v3.2.onnx	raw BGR 320x180	34.0%	31.1%	57.7%	52.4%	16.78	9.65	-1.47
2	3.4	SidewalkPilot-v3.4.onnx	raw BGR 320x180	33.3%	32.6%	60.7%	65.5%	14.38	4.86	0.80
3	3.1	SidewalkPilot-v3.1.onnx	raw BGR 320x180	28.1%	26.8%	53.3%	55.2%	22.65	9.73	-2.35
4	3.1b	SidewalkPilot-v3.1b.onnx	raw BGR 320x180	27.4%	25.8%	52.3%	56.7%	20.96	9.59	-3.77
5	3.3	SidewalkPilot-v3.3.onnx	raw BGR 320x180	27.1%	21.2%	48.1%	68.5%	14.63	5.20	-5.35
6	3.4b	SidewalkPilot-v3.4b.onnx	raw BGR 320x180	25.6%	22.7%	52.2%	73.7%	13.90	2.68	-2.09
7	3.2b	SidewalkPilot-v3.2b.onnx	raw BGR 320x180	25.1%	19.8%	46.0%	67.2%	14.46	5.91	-4.69
8	3.3b	SidewalkPilot-v3.3b.onnx	raw BGR 320x180	19.2%	10.6%	36.5%	78.0%	16.46	2.18	-8.29
9	3.0	SidewalkPilot-v3.0.onnx	raw BGR 320x180	16.0%	15.3%	42.9%	23.4%	18.97	13.62	-3.59
10	3.0b	SidewalkPilot-v3.0b.onnx	raw BGR 320x180	15.9%	14.8%	42.3%	24.6%	18.45	13.13	-3.94

Chronological Model Growth

S1/2 and S3 rows use different evaluation sets. Compare chronology and metrics only within the same eval-set group.

Model	Checkpoint filename	Series	Eval set	Prep	Scale	MAE	Median	Signed	<=2	<=5	<=20	Pred mean
1.0	SidewalkPilot-v1.0.pth	1	S1/2	raw BGR	86	25.55	18.46	-7.45	130	333	1184	88.87
1.0b	SidewalkPilot-v1.0b.pth	1	S1/2	raw BGR	86	25.07	17.45	-7.08	135	351	1227	89.24
1.1	SidewalkPilot-v1.1.pth	1	S1/2	raw BGR	86	20.85	13.42	-5.16	243	510	1416	91.16
1.1b	SidewalkPilot-v1.1b.pth	1	S1/2	raw BGR	86	20.85	13.42	-5.16	243	510	1416	91.16
1.2	SidewalkPilot-v1.2.pth	1	S1/2	raw BGR	86	18.64	11.79	-3.55	267	582	1519	92.77
1.2b	SidewalkPilot-v1.2b.pth	1	S1/2	raw BGR	86	18.63	11.85	-3.73	275	584	1547	92.59
1.3	SidewalkPilot-v1.3.pth	1	S1/2	raw BGR	86	18.55	11.81	-3.43	261	557	1545	92.89
1.3b	SidewalkPilot-v1.3b.pth	1	S1/2	raw BGR	86	18.48	11.24	-3.65	265	578	1553	92.67
1.4	SidewalkPilot-v1.4.pth	1	S1/2	raw BGR	86	16.81	10.19	-3.99	320	676	1620	92.33
1.4b	SidewalkPilot-v1.4b.pth	1	S1/2	raw BGR	86	17.04	10.07	-4.15	336	685	1598	92.17
1.5	SidewalkPilot-v1.5.pth	1	S1/2	raw BGR	86	16.82	10.59	-0.89	375	666	1572	95.43
1.5b	SidewalkPilot-v1.5b.pth	1	S1/2	raw BGR	86	16.82	10.59	-0.89	375	666	1572	95.43
1.6	SidewalkPilot-v1.6.pth	1	S1/2	raw BGR	86	17.53	11.21	0.21	316	630	1556	96.53
1.6b	SidewalkPilot-v1.6b.pth	1	S1/2	raw BGR	86	17.41	11.04	-0.01	318	629	1562	96.31
1.7	SidewalkPilot-v1.7.pth	1	S1/2	raw BGR	86	15.82	9.66	-0.48	415	744	1615	95.84
1.7b	SidewalkPilot-v1.7b.pth	1	S1/2	raw BGR	86	15.54	9.53	-0.87	410	769	1635	95.45
1.8	SidewalkPilot-v1.8.pth	1	S1/2	raw BGR	86	15.47	9.46	-0.65	478	796	1645	95.67
1.8b	SidewalkPilot-v1.8b.pth	1	S1/2	raw BGR	86	15.41	9.36	-0.70	492	794	1649	95.62
1.9	SidewalkPilot-v1.9.pth	1	S1/2	raw BGR	86	12.98	6.63	-2.06	584	966	1728	94.26
1.9b	SidewalkPilot-v1.9b.pth	1	S1/2	raw BGR	86	12.94	6.60	-2.06	597	979	1734	94.26
2.0	SidewalkPilot-v2.0.pth	2	S1/2	HSV/CLAHE -> BGR	85	11.12	5.09	-2.63	609	1068	1839	93.70
2.0b	SidewalkPilot-v2.0b.pth	2	S1/2	HSV/CLAHE -> BGR	85	11.13	5.07	-2.62	601	1064	1838	93.70
2.1	SidewalkPilot-v2.1.pth	2	S1/2	raw BGR	85	10.80	5.83	-1.15	477	982	1867	95.18
2.1b	SidewalkPilot-v2.1b.pth	2	S1/2	raw BGR	85	10.75	5.80	-1.12	480	978	1867	95.20
2.2	SidewalkPilot-v2.2.pth	2	S1/2	raw BGR	85	6.68	4.26	-0.22	594	1238	2094	96.10
2.2b	SidewalkPilot-v2.2b.pth	2	S1/2	raw BGR	85	6.68	4.23	-0.46	613	1255	2094	95.86
2.3	SidewalkPilot-v2.3.pth	2	S1/2	raw BGR	85	3.30	1.93	0.01	1129	1768	2194	96.33

Model	Checkpoint filename	Series	Eval set	Prep	Scale	MAE	Median	Signed	<=2	<=5	<=20	Pred mean
2.3b	SidewalkPilot-v2.3b.pth	2	S1/2	raw BGR	85	3.41	2.05	0.41	1088	1742	2195	96.73
2.4	SidewalkPilot-v2.4.pth	2	S1/2	raw BGR	85	3.28	2.46	-0.23	957	1724	2218	96.09
2.4b	SidewalkPilot-v2.4b.pth	2	S1/2	raw BGR	85	3.27	2.39	-0.04	953	1729	2218	96.28
3.0	SidewalkPilot-v3.0.onnx	3	S3	raw BGR 320x180	-	18.97	13.62	-3.59	6533	16039	53679	93.53
3.0b	SidewalkPilot-v3.0b.onnx	3	S3	raw BGR 320x180	-	18.45	13.13	-3.94	6762	16621	55133	93.18
3.1	SidewalkPilot-v3.1.onnx	3	S3	raw BGR 320x180	-	22.65	9.73	-2.35	28493	33498	50477	94.77
3.1b	SidewalkPilot-v3.1b.onnx	3	S3	raw BGR 320x180	-	20.96	9.59	-3.77	29744	34370	51732	93.35
3.2	SidewalkPilot-v3.2.onnx	3	S3	raw BGR 320x180	-	16.78	9.65	-1.47	28795	33405	55311	95.65
3.2b	SidewalkPilot-v3.2b.onnx	3	S3	raw BGR 320x180	-	14.46	5.91	-4.69	35019	39659	59670	92.43
3.3	SidewalkPilot-v3.3.onnx	3	S3	raw BGR 320x180	-	14.63	5.20	-5.35	35471	40392	58495	91.77
3.3b	SidewalkPilot-v3.3b.onnx	3	S3	raw BGR 320x180	-	16.46	2.18	-8.29	39671	43770	57789	88.83
3.4	SidewalkPilot-v3.4.onnx	3	S3	raw BGR 320x180	-	14.38	4.86	0.80	35152	40809	59702	97.92
3.4b	SidewalkPilot-v3.4b.onnx	3	S3	raw BGR 320x180	-	13.90	2.68	-2.09	38934	43481	60361	95.03

Series 1/2 Field-Case / Subset MAE

Lower is better. All rows in this table use the same evaluation dataset.

Model	D0328_17	D0329_15	D0425_14	D0426_18	D0427_18	D0429_19	D0502_12	D0502_19	D0503_17	D0506_20	D0510_18	D0510_19	D0510_20
2.3b	2.45	2.57	3.41	3.15	3.42	2.52	3.46	3.23	2.50	4.00	4.47	2.79	4.60
2.4b	3.16	2.92	4.11	4.04	4.08	2.51	4.62	3.58	2.78	2.47	3.13	1.83	3.15
2.3	2.39	2.51	3.30	3.04	3.46	2.54	3.23	3.07	2.38	3.71	4.20	2.65	4.50
2.4	3.14	2.96	4.09	3.97	4.12	2.48	4.66	3.60	2.84	2.48	3.21	1.69	3.16
2.2	3.84	3.62	5.42	4.70	5.66	4.39	5.68	4.74	3.53	2.50	14.71	8.43	10.51
2.2b	3.81	3.56	5.46	4.97	5.61	4.38	5.67	4.72	3.50	2.62	14.65	8.25	10.58
2.1	19.70	20.13	3.69	3.26	3.89	3.32	4.14	3.62	2.47	3.60	10.64	6.88	8.76
2.0b	18.09	20.54	2.69	2.02	2.57	1.70	2.32	2.25	1.66	35.80	11.71	7.49	10.54
2.1b	19.65	19.96	3.65	3.19	3.81	3.33	4.10	3.64	2.42	3.63	10.67	6.86	8.73
2.0	17.97	20.51	2.71	2.03	2.57	1.69	2.35	2.26	1.66	36.00	11.70	7.45	10.59
1.9b	17.10	21.79	1.67	1.49	1.73	1.43	1.36	1.78	28.24	35.89	10.51	5.75	10.87
1.9	17.27	21.89	1.66	1.52	1.74	1.41	1.32	1.79	28.23	35.45	10.48	5.70	10.91
1.8b	17.43	20.92	1.52	1.19	1.42	0.97	1.08	22.57	28.02	37.04	12.89	5.38	14.69
1.8	17.40	20.92	1.51	1.18	1.42	0.96	1.08	22.62	28.08	37.19	13.08	5.39	14.87
1.7b	16.52	20.73	1.39	1.03	1.23	1.01	17.43	22.69	27.59	30.87	10.19	5.57	12.68
1.7	16.55	21.03	1.44	1.12	1.23	0.93	17.51	22.84	27.33	30.78	10.21	5.76	13.50
1.5	16.69	20.32	1.24	0.88	0.99	25.74	18.59	21.31	26.42	26.32	10.75	6.48	15.96
1.5b	16.69	20.32	1.24	0.88	0.99	25.74	18.59	21.31	26.42	26.32	10.75	6.48	15.96
1.4	16.34	20.23	0.88	0.56	31.68	25.83	18.73	24.90	26.43	34.05	10.71	6.82	11.14
1.6b	17.01	20.01	1.24	0.90	1.10	26.22	19.02	22.90	27.36	29.09	10.00	5.66	17.51
1.6	16.84	20.09	1.26	0.84	1.10	26.32	18.90	22.95	27.36	29.10	9.99	5.64	17.99
1.4b	16.94	20.62	1.00	0.72	31.91	26.65	19.21	24.99	26.13	34.16	10.42	7.00	11.30
1.2	16.38	19.88	0.87	39.03	31.76	25.39	20.86	23.31	32.51	36.88	14.69	6.91	11.84
1.2b	16.39	19.86	0.93	38.42	32.11	25.46	21.01	23.36	32.64	36.64	14.57	6.80	11.77
1.3	16.46	20.12	0.87	37.56	34.06	25.27	19.51	23.50	34.04	32.99	13.71	6.20	11.46
1.3b	16.63	20.54	1.02	36.75	35.63	24.12	19.66	23.53	34.63	32.20	13.01	6.25	10.81
1.1	17.12	20.33	1.32	41.55	36.60	33.44	21.42	26.71	33.09	29.72	13.52	5.50	17.34

Model	D0328_17	D0329_15	D0425_14	D0426_18	D0427_18	D0429_19	D0502_12	D0502_19	D0503_17	D0506_20	D0510_18	D0510_19	D0510_20
1.1b	17.12	20.33	1.32	41.55	36.60	33.44	21.42	26.71	33.09	29.72	13.52	5.50	17.34
1.0b	17.36	20.30	49.19	45.40	54.06	37.39	29.15	30.30	40.51	40.61	17.89	15.52	18.26
1.0	18.07	21.08	49.02	46.31	53.69	37.44	29.56	30.14	41.72	40.76	17.95	14.47	18.74

Series 3 Field-Case / Subset MAE

Lower is better. All rows in this table use the same evaluation dataset.

Model	D0702_16	D0707_16	D0712_15	D0712_16
3.2	14.46	13.64	23.37	22.43
3.4	11.69	13.02	17.00	20.73
3.1	16.00	23.15	35.18	36.67
3.1b	14.18	21.84	33.49	35.17
3.3	11.18	13.84	24.05	21.94
3.4b	11.28	15.94	17.65	19.02
3.2b	11.70	14.37	22.00	20.13
3.3b	11.27	16.20	23.50	27.85
3.0	17.49	16.97	24.27	22.49
3.0b	16.85	17.06	24.90	21.94

Prediction Distribution

Model	Pred min	P05	P25	Median	P75	P95	Pred max	Pred mean	Target mean
2.3b	5.00	58.64	83.54	93.98	104.04	164.76	175.00	96.73	96.32
2.4b	5.00	57.77	82.46	93.29	103.57	164.71	175.00	96.28	96.32
2.3	5.00	58.42	83.26	93.64	103.24	165.00	175.00	96.33	96.32
2.4	5.00	57.47	82.19	93.08	103.48	164.71	175.00	96.09	96.32
2.2	5.00	51.21	81.67	93.65	105.09	167.06	175.00	96.10	96.32
2.2b	5.00	51.20	81.50	93.35	104.86	167.64	175.00	95.86	96.32
2.1	5.00	46.92	79.86	93.56	105.96	161.84	175.00	95.18	96.32
2.0b	5.00	41.26	77.43	92.44	106.38	160.18	175.00	93.70	96.32
2.1b	5.00	47.17	79.95	93.49	105.85	161.30	175.00	95.20	96.32
2.0	5.00	41.35	77.37	92.45	106.30	160.31	175.00	93.70	96.32
1.9b	4.00	46.67	78.77	92.67	107.24	157.44	176.00	94.26	96.32
1.9	4.00	46.17	78.77	92.52	107.33	157.92	176.00	94.26	96.32
1.8b	4.00	48.10	80.15	94.64	109.16	154.91	176.00	95.62	96.32
1.8	4.00	48.29	80.12	94.60	109.19	155.61	176.00	95.67	96.32
1.7b	4.00	49.29	81.17	94.08	108.57	153.13	176.00	95.45	96.32
1.7	4.00	49.39	81.31	94.44	109.12	153.53	176.00	95.84	96.32
1.5	4.00	47.33	80.30	93.99	108.59	153.09	176.00	95.43	96.32
1.5b	4.00	47.33	80.30	93.99	108.59	153.09	176.00	95.43	96.32
1.4	4.00	48.35	80.64	92.55	102.58	136.88	176.00	92.33	96.32
1.6b	4.00	48.23	79.75	94.62	109.73	154.52	176.00	96.31	96.32
1.6	4.00	48.57	80.01	94.65	109.88	154.94	176.00	96.53	96.32
1.4b	4.00	48.10	80.18	92.30	101.93	137.30	176.00	92.17	96.32
1.2	4.00	47.80	78.33	93.44	107.11	136.20	176.00	92.77	96.32
1.2b	4.00	47.73	78.53	93.35	106.81	137.56	176.00	92.59	96.32
1.3	4.00	47.72	79.07	93.19	107.25	133.68	176.00	92.89	96.32
1.3b	4.00	47.97	79.35	92.96	106.07	134.30	176.00	92.67	96.32
1.1	4.00	44.02	75.57	90.82	106.24	141.57	176.00	91.16	96.32
1.1b	4.00	44.02	75.57	90.82	106.24	141.57	176.00	91.16	96.32

Model	Pred min	P05	P25	Median	P75	P95	Pred max	Pred mean	Target mean
1.0b	9.01	43.91	72.06	88.65	104.89	136.78	173.93	89.24	96.32
1.0	8.46	42.83	71.11	88.86	105.18	136.92	174.15	88.87	96.32
3.2	0.30	67.36	80.51	90.15	100.10	155.73	179.83	95.65	97.12
3.4	1.36	69.15	88.74	90.13	99.63	151.84	179.54	97.92	97.12
3.1	0.74	19.83	87.42	89.79	111.08	159.16	179.25	94.77	97.12
3.1b	0.95	21.03	87.67	89.94	100.39	157.76	179.06	93.35	97.12
3.3	0.66	67.78	88.57	90.10	91.51	151.77	178.73	91.77	97.12
3.4b	5.13	68.57	88.86	89.97	91.27	149.68	178.76	95.03	97.12
3.2b	0.96	68.22	88.79	90.21	91.88	129.43	178.79	92.43	97.12
3.3b	5.27	26.21	88.79	89.87	91.17	111.87	175.51	88.83	97.12
3.0	9.30	65.26	81.73	93.48	104.93	122.53	172.37	93.53	97.12
3.0b	12.34	66.28	82.19	93.31	104.22	119.72	170.13	93.18	97.12

Prediction Buckets vs Ground

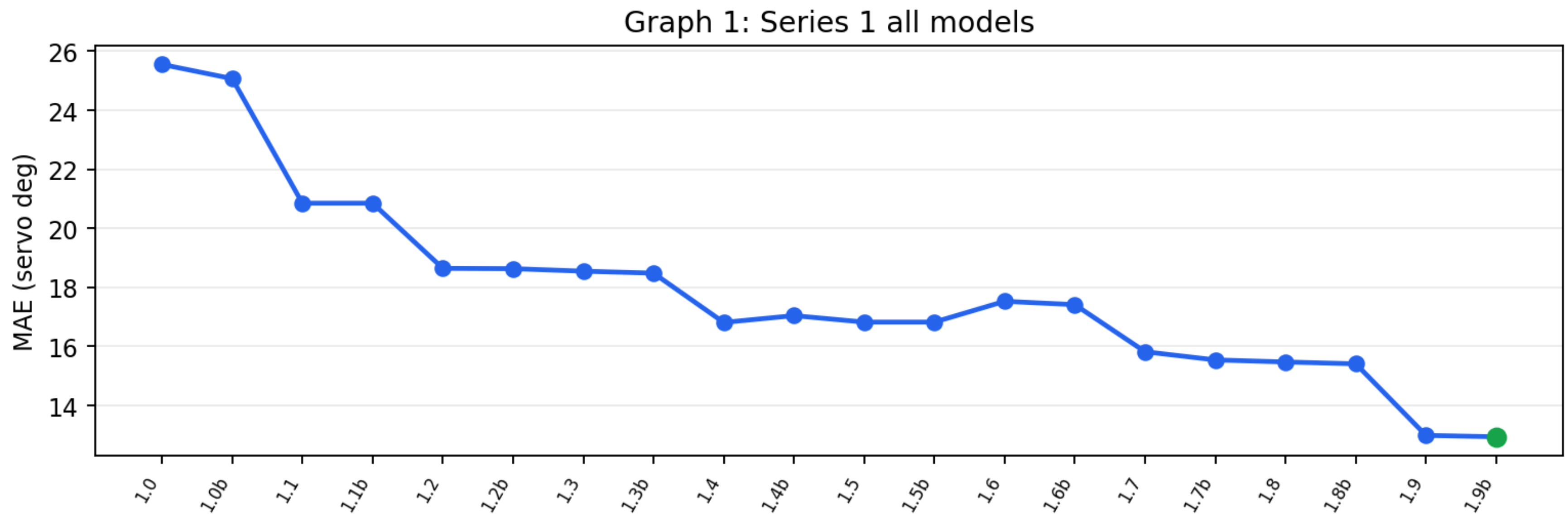
Bucket columns show prediction counts; exact/off-by-one compare prediction bucket against ground bucket.

Model	HL 0-45	L 45-75	SL 75-85	ST 85-95	SR 95-105	R 105-135	HR 135-180	Exact	Off by <=1
2.3b	74	173	372	559	531	307	208	75.7%	98.2%
2.4b	76	199	391	543	507	296	212	71.2%	98.5%
2.3	73	181	380	578	514	293	205	74.3%	98.2%
2.4	76	209	389	552	498	292	208	70.8%	98.4%
2.2	94	244	366	488	473	342	217	58.9%	92.3%
2.2b	94	252	371	493	465	330	219	58.6%	92.0%
2.1	97	331	303	456	455	358	224	50.7%	84.8%
2.0b	123	373	310	432	394	373	219	49.7%	84.7%
2.1b	96	327	309	449	458	361	224	50.2%	84.9%
2.0	123	375	310	430	395	374	217	49.3%	84.8%
1.9b	107	352	320	453	387	400	205	44.0%	81.9%
1.9	106	354	322	451	388	399	204	43.4%	81.8%
1.8b	99	332	278	426	404	481	204	40.0%	78.6%

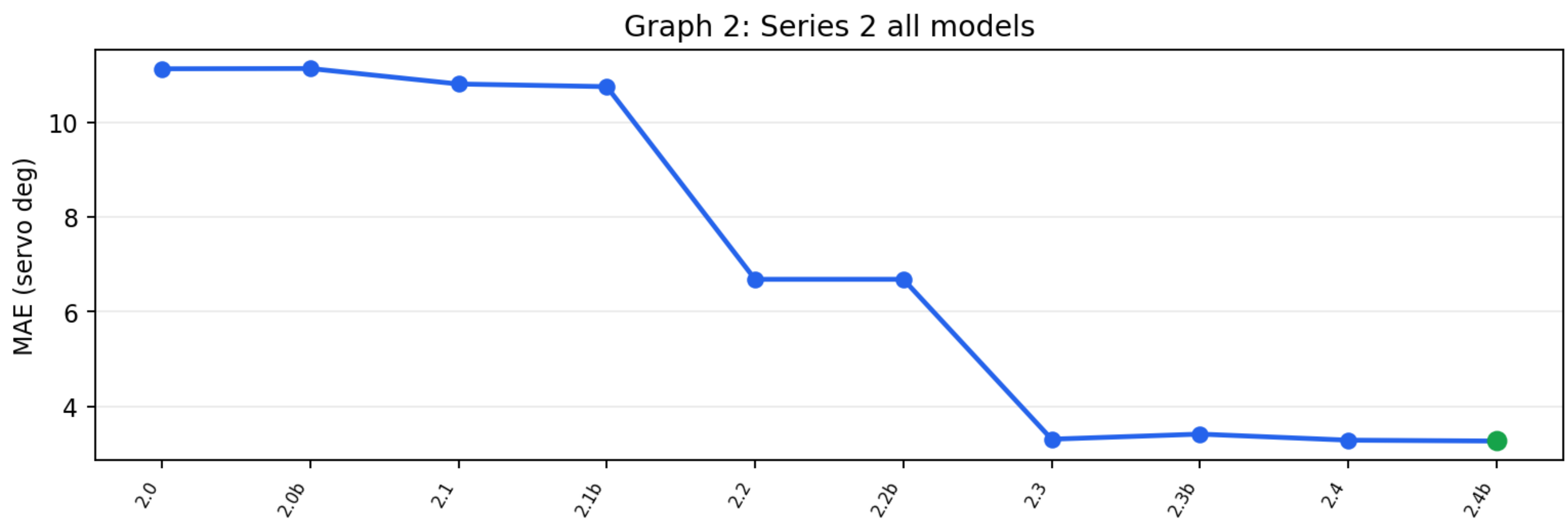
Model	HL 0-45	L 45-75	SL 75-85	ST 85-95	SR 95-105	R 105-135	HR 135-180	Exact	Off by <=1
1.8	99	332	276	424	404	479	210	40.1%	78.6%
1.7b	92	313	280	470	418	456	195	39.3%	79.6%
1.7	93	313	271	463	403	477	204	39.1%	78.9%
1.5	99	315	304	446	393	465	202	35.1%	76.9%
1.5b	99	315	304	446	393	465	202	35.1%	76.9%
1.4	86	308	326	545	482	357	120	34.4%	75.4%
1.6b	92	331	290	420	413	440	238	34.2%	76.8%
1.6	91	339	282	420	405	445	242	33.5%	76.5%
1.4b	94	300	313	589	470	333	125	34.5%	75.0%
1.2	92	382	282	428	424	497	119	32.6%	74.4%
1.2b	93	394	273	433	425	483	123	31.7%	74.3%
1.3	88	365	310	432	398	523	108	30.8%	73.5%
1.3b	85	342	322	455	421	491	108	31.8%	74.0%
1.1	119	425	326	420	344	449	141	27.7%	68.4%
1.1b	119	425	326	420	344	449	141	27.7%	68.4%
1.0b	121	530	308	382	329	432	122	21.1%	58.2%
1.0	134	528	327	345	328	440	122	19.6%	56.2%
3.2	1947	8571	10447	34948	7681	8398	9245	46.9%	72.1%
3.4	647	4163	10475	44075	2656	9437	9784	56.3%	77.2%
3.1	7128	4482	8105	37535	3366	8513	12108	47.1%	67.1%
3.1b	6496	4504	8586	38860	2793	10380	9618	48.1%	68.2%
3.3	1649	7881	7052	48527	5328	6101	4699	54.0%	74.1%
3.4b	548	4956	8847	52257	363	7193	7073	58.2%	76.7%
3.2b	2031	3326	10037	47739	6393	7651	4060	52.8%	76.4%
3.3b	5114	2745	2420	58965	4711	4430	2852	56.2%	72.2%
3.0	325	10946	14167	18078	17519	19277	925	23.0%	61.7%
3.0b	265	10415	14295	18938	18306	18384	634	23.5%	63.1%

Graphs

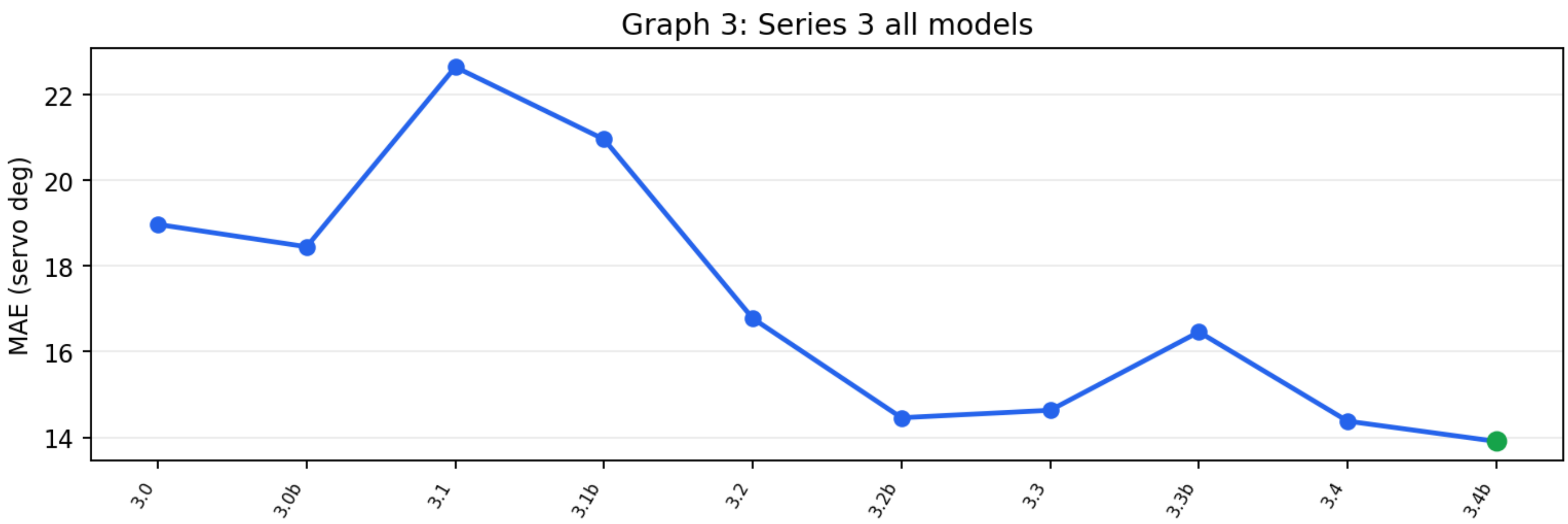
Graph 1: Series 1 all models



Graph 2: Series 2 all models



Graph 3: Series 3 all models



Steering-Bucket Confusion (9-class) — latest 4 models

Rows = the TRUE steering bucket, columns = the model's PREDICTED bucket. Green diagonal = correct; red off-diagonal = confusion. Shading = share of that true bucket (each row sums to ~100%). Buckets: HL 0-45, L 45-60, L+ 60-75, SL 75-85, ST 85-95, SR 95-105, R 105-120, R+ 120-135, HR 135-180.

v3.4b · exact bucket 57% · within one bucket 74%

T\P	HL	L	L+	SL	ST	SR	R	R+	HR
HL	146	34	220	81	152	1	2	2	53
L	28	36	362	218	363	1	2	2	15
L+	28	47	1043	671	842	3	6	9	34
SL	43	49	817	1510	1680	2	11	29	58
ST	133	120	2061	5852	39710	160	1340	1588	2946
SR	6	1	27	114	1746	52	259	246	344
R	6	1	23	138	2315	33	461	397	525
R+	8	·	32	77	2068	41	557	569	709
HR	150	5	78	186	3381	70	986	727	2389

v3.4 · exact bucket 54% · within one bucket 74%

T\P	HL	L	L+	SL	ST	SR	R	R+	HR
HL	260	27	155	109	129	1	3	1	6
L	19	85	374	243	289	3	4	·	10
L+	21	27	1089	860	645	11	7	4	19
SL	33	10	643	2208	1223	19	21	12	30
ST	144	67	1636	6739	35306	1283	2867	1467	4401
SR	1	·	11	72	1301	344	408	239	419
R	4	·	6	69	1557	275	905	373	710
R+	5	·	4	42	1336	311	711	746	906
HR	160	·	29	133	2289	409	1149	520	3283

v3.3b · exact bucket 55% · within one bucket 70%

T\P	HL	L	L+	SL	ST	SR	R	R+	HR
HL	245	·	110	40	244	19	14	·	19
L	176	1	154	66	573	40	12	·	5
L+	427	3	440	217	1413	128	36	·	19

T\P	HL	L	L+	SL	ST	SR	R	R+	HR
SL	440	2	387	389	2669	211	70	1	30
ST	3071	18	1326	1457	42065	3051	1820	99	1003
SR	111	·	40	44	1961	219	288	10	122
R	146	·	56	51	2726	280	418	19	203
R+	145	·	56	41	2604	294	565	40	316
HR	353	·	152	115	4710	469	979	59	1135

v3.3 · exact bucket 53% · within one bucket 71%

T\P	HL	L	L+	SL	ST	SR	R	R+	HR
HL	300	34	132	42	156	15	6	·	6
L	76	67	347	109	370	32	10	3	13
L+	136	90	1010	431	853	102	50	4	7
SL	149	120	844	1205	1587	182	99	4	9
ST	757	795	3502	4578	36908	3256	2239	349	1526
SR	20	17	108	117	1596	333	307	92	205
R	28	49	181	144	2066	375	518	187	351
R+	30	23	162	134	1878	387	562	337	548
HR	153	60	340	292	3113	646	986	348	2034

Notes

- Series 1/2 checkpoints were evaluated on 2,224 original corrected real images. Series 3 checkpoints were evaluated on 81,237 current Series 3 images. Cross-dataset metric values are not directly comparable.

- Series 2 v2.0/v2.0b used their required HSV/CLAHE preprocessing; all other Series 1/2 models used raw BGR.

- MAE and within-degree counts can reward straight collapse. Use the class-balanced and turn-recall columns for selection, with MAE as supporting evidence.

- Offline MAE does not prove real-world reliability. The car can still fail on lighting, turns, driveways, road-edge ambiguity, speed, and sensor conditions.

- Series 1/2 training was CARLA-assisted, but this report follows the historical evaluator and scores them on the 2,224 corrected real-image set rather than synthetic training frames.

- Series 3 models were trained on overlapping images from this dataset, so their scores are fit checks, not held-out generalization estimates.

- The Series 3 evaluation set contains 81,237 curated real field frames across five manual-driving runs from July 2 through July 12, 2026.

- Series 3 throttle labels are near-constant at full throttle, so this report evaluates steering only; throttle control remains disabled pending varied-throttle data.

- v3.3, v3.3b, v3.4, and v3.4b still require field testing before any deployment winner is declared.